

DATA MODEL

26 TABLES LIVE ON LOCAL POSTGRES 16

core.entity

PK entity_id = CIK
natural key, not a uuid

md.security

PK figi · FK cik
ticker is a label, not a key

md.quote_snapshot

FK figi · immutable
source · delay · as_of

entity_id

entity_id = cik

figi

core.entity_synonym

kind · confidence
22 NVDA variants seeded

pos.position_snapshot

FK figi · as_of date
Orion, reconciled T-1

edgar.filing

PK accession_no
the citation key

ops.advisor_interaction

channel = internal_chat
compliance join key

news.candidate_link

asserted = false (CHECK)
never a stated cause

news.headline

allowlisted publishers
url · published_at

ops.flag

supervision writes only
desk cannot clear its own

ops.hitl_queue

every ASK / BLOCK
escalation is an outcome

ops.failure_log

retry exhaustion
no substituted default

WORKFLOW

GATE FIRST, COMMIT ONCE

0

gate_agent

— PROCEED / ASK / BLOCK before any data is touched.
Zero data scope by design.

1

advisor_desk_orchestrator

parses intent, writes the interaction row, routes.

2

market_clock

— outside the NYSE regular session no call is made at all. Two thirds of a 24-hour schedule, free.

3

quote_snapshot_agent

(no model) fetches, re-checks once on empty, compares two reads, escalates on a halt.

4

research_orchestrator

→ filing_fact + macro_news, in parallel, returning rows only.

5

discrepancy_agent

when two sources disagree — reports the divergence, never picks.

6

heuristic_override_agent

runs last; every fire writes rule_id + before/after.

7

advisor_answer_agent

composes from returned rows. It cannot fetch and cannot send.

8

supervision_orchestrator

holds the only write on ops.flag. The desk cannot clear a flag on its own output.

STOOD UP

261460

TABLESTESTS GREENMEASURED EVALS

- Postgres 16, schemas 001–006 applied
- CIK canonical ids, CHECK-enforced by entity type
- One money rule: numeric(20,6), Rule 612 on quotations
- Per-agent roles + RLS failing closed
- NYSE clock with the 2026 holiday table
- Call budget: session TTL, dedupe, monthly pacing
- Pydantic at every ingest boundary
- CCO halt notification path, transport-injected
- Advisor desk front end: 502-symbol resolver, live EDGAR citations

Not stood up: no agent runtime, no live pull (egress blocked), no auth, no deployment.

RED TEAM

12 DEFECTS FOUND AND FIXED AFTER THE BUILD WAS “DONE” · 8 NEW REGRESSION TESTS

Every one of these passed 138 green tests. They survived because the fixtures were hand-written with clean values and egress is blocked, so **no real payload ever crossed the boundary**. Two would have taken the desk down on day one. This is the argument for a red-team pass as a build step rather than a phase: a green suite proves the code does what the tests say, not what the source does.

ID	DEFECT	WHAT IT WOULD HAVE COST IN PRODUCTION
RT-01	Every real price failed validation. decimal_places=6 is a <i>maximum</i> in pydantic; Yahoo serializes doubles, so 226.03999328613281 raised.	Total outage of the price path, presenting as “source unavailable.” Days lost blaming the vendor.
RT-02	The NYSE calendar was bounded only above. Any date before 2026 scored against the 2026 holiday table: 2025-12-25 and 2024-07-04 both returned <i>regular session</i> .	A backfill or fixture replay marked tradeable on Christmas. Silent, and wrong in the direction that fetches.
RT-11	The poller exited 0 on failure. main() compared status to the literal “FAILED”, so FAILED_NO_DATA, both INDETERMINATE states and FALLBACK_REQUIRED all reported success.	Cron logs green over a desk serving nothing. The failure mode CLAUDE.md #6 exists to prevent, in the monitoring layer.
RT-03/04	Only HTTPError was caught. A reset or timeout on the EDGAR leg propagated out, killed the price leg, and skipped save_state() — losing the ETag and the call counters with it.	One network blip burns the conditional-GET cache and double-charges the monthly cap.
RT-07	A cache hit returned no row. The dedupe that justifies the entire call budget delivered {"status": "cache"} and nothing else.	The budget design was decorative: N advisors, N calls.
RT-05/06	The audit record lied. The exception was discarded, so a parser bug and a vendor outage wrote identical failure rows; the recovery path wrote “verified by a second read” when nothing was verified.	Un-triageable incidents, and an audit trail that misstates what happened – the worst kind, because it is confident.
RT-12	A clean checkout could not run. make setup installed jsonschema (imported nowhere) and not pydantic (imported everywhere).	Every new engineer’s first hour. Now pinned in requirements.txt.
RT-13	Thirteen S&P 500 tickers are English words. The front-end resolver read “Why is <i>it</i> down today?” as Gartner (IT) and “what is that worth right <i>now</i> ?” as ServiceNow (NOW).	The precise failure this system exists to prevent: a confident, sourced, fully-cited answer about the wrong company. Found by running the page headless, not by reading it.

Excluded from this list as boundary conditions, not defects: blocked egress, the stub eval adapter, the absent entitled feed, and the absent agent runtime. Those are known and scoped. Also fixed but not tabled: RT-09 (the documented cron and make poll both used a 15-minute interval that the budget refuses to start on) and RT-10 (retrieved_at stamped at the top of the cycle, so a slow EDGAR call made a fresh print fail as “a quote from the future”).

EVALUATION

49 GOLDEN CASES · 25 CATEGORIES · ADAPTER = STUB

WRITTEN

49 cases · gate block · gate ASK · **gate persistence under pressure** · ceiling rule · least privilege · fail-loud · no-send · no-fetch · citation sufficiency · silo integrity · source disagreement · trading halt · market closed · provenance · Rule 612 precision · override fires *and* declines · no-causation · unavailable field. Every agent and every live orchestrator has at least one case.

NOT WRITTEN

no measured numbers

The adapter is a stub, so nothing scores the real system and no baseline is set — setting one from stub mode would be a false record. Also absent: latency p95 · cost per task from observed tokens · prompt-injection through a filing or headline · multi-turn drift within a session · entity-resolution precision and recall against a labeled set · XBRL extraction fidelity vs. ground truth · any symbol other than NVDA.

Front end: the desk UI has no test at all. The homograph defect (RT-13) was found by a one-off headless run, not a suite. A Playwright case per intent is the cheapest eval left unbuilt.

NEEDED BEFORE PRODUCTION

fidelity

Citation faithfulness — every numeric claim must resolve to a row containing it. Deterministic, not LLM-judged.

calibration

Over-refusal rate measured beside refusal precision. A gate that blocks everything scores 100% on safety and is useless.

drift

Frozen-fixture replay weekly: identical input, changed output *is* drift. Per-case comparison, not aggregate — the config already does this.

drift

Model-swap regression across the routing tiers, to price the Sonnet/Opus bet with data instead of argument.

human

Supervisory sampling — N% of turns to a principal with a recorded agreement rate. This is the artifact a Rule 3110 examination asks for, and no vendor can produce it for you.

scale

≥200 cases across ≥25 symbols, including halted, sub-\$1, and thin-book names.

BURNDOWN TO PRODUCTION

RANKED, DEPENDENCIES FIRST

1

Agent runtime

— implement SystemAdapter.invoke and the six domain agents. Until this lands, nothing is measured.

2

Entitled market-data feed

— delete the interim source path; bid/ask/spread and halt status arrive with it.

3

Orion + clearing reads

— intraday quantity, so a notional stops mixing T-1 shares with a live price.

4

Retention and audit pipeline

— raw turn persisted before compaction, immutable, exportable.

5

WSP rule table + supervision agent

— the flag endpoints have no referent until the rules have ids.

6

AuthN/AuthZ wiring

— the RLS boundary is only real if the runtime connects as the agent's role.

7

Eval corpus + first real baseline

, then drift on every deploy.

8

Observability

— failure_log and hitl_queue triage in front of a human, not in a table.

9

Scale to a watchlist

and re-measure cost: the cap scales linearly with symbol count.

10

HA, backup, key management, DR.

11

Front-end test suite

— headless per-intent assertions, so a resolver change cannot silently answer about the wrong issuer again.

GATING ISSUES NO THIRD PARTY CAN RESOLVE			BROKER MANAGEMENT OWNS THESE
DECISION	WHY A VENDOR CANNOT MAKE IT	BLOCKED UNTIL DECIDED	
Market-data entitlement	An exchange agreement with per-user professional-subscriber reporting is a contract the firm signs, and Rule 603(c) governs how a quote may be displayed.	real-time quotes; the whole bid/ask feature	
Is desk output a recommendation?	Reg BI turns on the firm's own view of what its reps do with an answer. Nobody outside can classify it.	any advisor-facing rollout	
WSPs as testable rules	The procedures are the firm's, in prose, with firm-specific ids and owners. Converting them is a compliance project, not an integration.	every flag endpoint; supervision silo	
Supervisory review model	Who reviews, at what sampling rate, with what escalation and what evidence — Rule 3110 asks the firm, not the tool.	human-review evals; examination readiness	
Authoritative source for intraday quantity	Orion vs. the clearing firm is a books-and-records decision with a supervisory owner.	notional; unrealized P/L	
Retention design for AI turns	What constitutes the record, and for how long, is the firm's counsel's call.	compaction economics; audit export	
Provisional-match tolerance	Whether 0.90 confidence is enough to act on a resolved name is a risk-appetite question.	entity-resolution thresholds	
Model-use governance	Which models, in which region, over which data, with what vendor terms.	routing; any client-data path	
Cost cap and cadence per advisor	A spend ceiling is a budget decision that changes the architecture, not a tuning knob.	polling design; watchlist scale	

Honest state. No live pull has run — this environment's egress policy returns 403 for sec.gov and finance.yahoo.com, so the poller is proven against recorded payloads only. The eval harness runs end to end in **stub** mode, which is why no pass rate appears anywhere on this page. The price source and the SEC form allowlist are hardcoded pending client decisions, recorded in ADR-0010 and ADR-0011.



Ask the desk

Answers cite their source, or they escalate.

☐ audit trail

YOU

What is NVDA trading at right now?

DSK

NVDA · NVIDIA Corporation · CIK 0001045810 — the symbol pinned above the fold

ANSWER

NVDA 226.040000 USD, down 4.32 (−1.88%) against a prior close of 230.360000.

This is **15 minutes delayed**. It is not the price you would trade at, and the halt state is unverified because this source does not report one. Bid, ask and spread are absent for the same reason — they need the entitled feed.

source yahoo_chart_v8 · delayed 900s · as of Sep 8, 01:30 PM ET · basis adjusted · halt unverified

YOU

Why is it down today?

DSK

NVDA · NVIDIA Corporation · CIK 0001045810 — the symbol pinned above the fold

No security named, so I am reading this against the symbol pinned above — **NVDA**, NVIDIA Corporation, CIK 0001045810. Load a different symbol and the same question resolves to that one.

MOVE: ANSWERED · CAUSE: UNKNOWN

Yes — **NVIDIA Corporation (NVDA) is down**. Last **226.040000** against a prior close of 230.360000: **−4.32 (−1.88%)**, as of Sep 8, 01:30 PM ET on a 15-minute delay.

Why: not established. I can tell you what the move was, because that is arithmetic on two rows I hold. I cannot tell you what caused it, and I will not narrate one. The nearest disclosure in the window is a **Form 4** accepted at Sep 8, 12:31 PM ET — 0001045810-26-000110 — which is *temporally adjacent and nothing more*. A Section 16 report of an insider transaction is not an explanation of a −1.88% index-constituent move.

Headline candidates are surfaced with the basis for each link — *issuer named, sector named, geography named* — and never as a cause. Promotion to a cause is blocked by the system, not merely discouraged: nothing here can label a headline an explanation. Ranking one into an explanation needs a relevancy model, which is out of scope for this prototype.

What I would need to answer "why" honestly: an intraday tick series (entitled feed), a headline corpus with timestamps. and a ranking model with a measured precision floor. Absent those. "I do

NVDA

NVIDIA Corporation

CIK 0001045810 · FIGI BBG000BBJQV0 · XNGS

LOAD A SYMBOL

Load

SPREAD

undefined

Blocked on the entitled feed. No book, so no mid. Never derived from OHLC.

502 S&P 500 constituents resolve by ticker → name → CIK. Only **NVDA** carries a seeded price slice.

market open

delayed 15 min — not a tradeable price

halt status unverified

as of Sep 8, 01:30 PM ET

no bid/ask — entitled feed required

BID / ASK

not available

Blocked on the entitled feed. This source has no quote book.

name — you always see the canonical record.

Type any of them into the box and the desk answers about NVIDIA Corporation — but it answers as NVIDIA Corporation, and below 0.90 confidence it tells you it made a provisional match rather than applying one silently.

On file: 8 misspellings, 2 short names, 1 legal name, 1 ticker, 1 exchange qualified ticker, 1 phonetic. Of those, **9** sit below the 0.90 confidence line, so the desk tells you it made a provisional match instead of applying one silently.

Deliberately **absent**: TRT:NVDA, the Toronto-listed CAD-hedged CDR. It shares the ticker and is a different instrument. A synonym table that swallows it is worse than no synonym table.

source firm security master · canonical id CIK 0001045810

YOU

Any recent NVDA filings?

DSK

NVDA · NVIDIA Corporation · CIK 0001045810 — the symbol pinned above the fold

ANSWER

Three NVDA filings on the intraday-material allowlist — each accession links to the filing itself on EDGAR:

4 — Section 16 insider transaction, filed Sep 8, 12:31 PM ET 0001045810-26-000110

SC 13G/A — Beneficial ownership amendment, filed Sep 4, 02:02 PM ET 0001045810-26-000109

13F-HR — Institutional holdings, quarterly, filed Aug 14, 11:00 AM ET 0001045810-26-000101

10-K and 10-Q are excluded by design: they land outside trading hours. Form 13H is on the allowlist but will never appear — large-trader filings are confidential under Exchange Act 13(h)(7).

source SEC EDGAR · issuer CIK 0001045810

What is NVDA trading at right now?

Give me the bid and ask on NVDA

Why is it down today?

What did the Wilson NVDA position close at?

What's that position worth right now?

Any recent NVDA filings?

Resolve a fat-fingered name to its CIK

What is AMD trading at?

Should my client buy more NVDA?

Draft this as an email to my client

e.g. what is NVDA trading at right now?

Ask

Book of record		
Orion — positions		T - 1
Orion Eclipse — trading		STUB
Redtail — households		CRM
Pershing NetX360 — clearing		STUB
Clearing statements		STUB
Stubbed pending client guidance. Every link is a placeholder; none of these systems is connected. Which of them are in scope, what each returns, and whether Orion stays read-only are client decisions — the panel is built to be filled in from the firm's own system inventory and entitlements, not guessed at here.		

Supervision		
Written Supervisory Procedures		WSP
Rule 3110 review queue		STUB
Communications archive		17A - 4
Advertising review — Rule 2210		STUB
Outside business activity		STUB
Gifts & entertainment log		STUB
Stubbed pending client guidance. The flag endpoints are built; the rules behind them are deliberately empty, so nothing fires today. The firm's actual WSP text, its 3110 review thresholds and its escalation path are client inputs : a supervisory rule invented by us is a compliance liability, not a feature.		

Service desk		
Stubbed pending client guidance. Destinations, SSO behaviour and which resources an advisor sees by role are configuration the firm supplies. Each item here is a slot, sized and placed, waiting for a real URL.		

NVDA

NVIDIA Corporation
CIK [0001045810](#) · FIGI [BBG000BBJQV0](#) · XNGS

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| source firm security master · canonical id [CIK 0001045810](#)

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